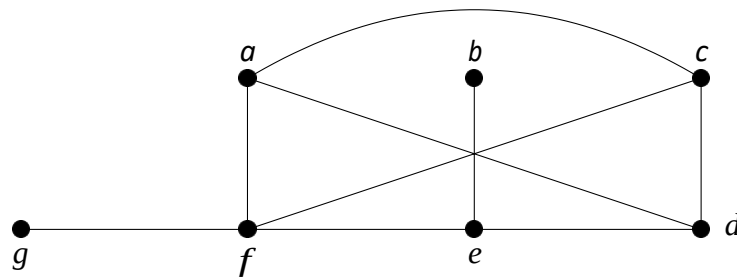


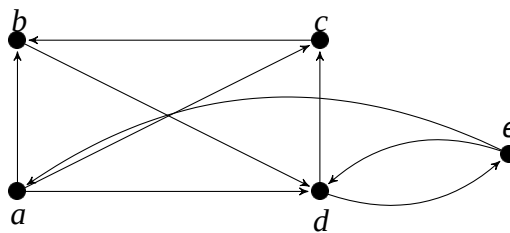
F.Y.B.Sc Data Science Sem II
GRAPH THEORY PRACTICALS

Practical no. 1: Introduction to graphs-I

1. Draw an undirected graph on following set of vertices and edges using Maxima software.
 $V = \{a, b, c, d, e\}, \quad E = \{(a, b), (a, d), (b, c), (d, b), (c, d), (e, a), (e, b), (e, c)\}.$
2. Find the number of vertices, the degree of each vertex, minimum and maximum degree vertex in the given undirected graph.



3. Find the in-degree and out-degree of each vertex in the digraph G with directed edges shown in following figure.



4. Draw the following graphs using Maxima software.

$$i) K_5$$
$$ii) C_6$$

iii) W_7

 $iv) K_{4,5}$

Practical no. 2: Introduction to graphs-II

1. Represent each of these graphs with an adjacency matrix:

i) K_5

ii) $K_{3,4}$

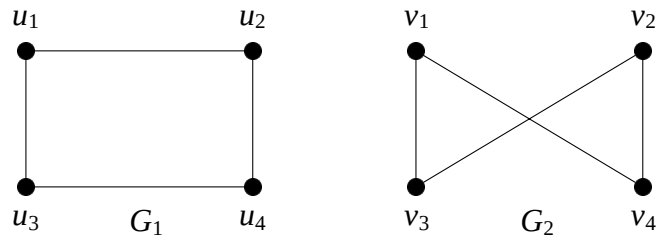
iii) W_6

2. Draw a graph represented by adjacency matrix given below.

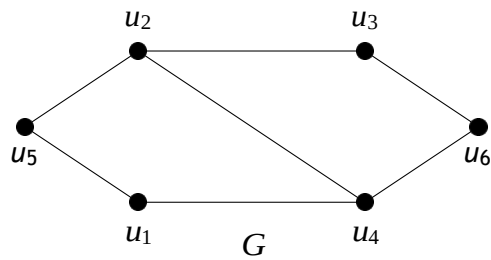
i)
$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

ii)
$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

3. Show that the following two graphs are isomorphic using Maxima.



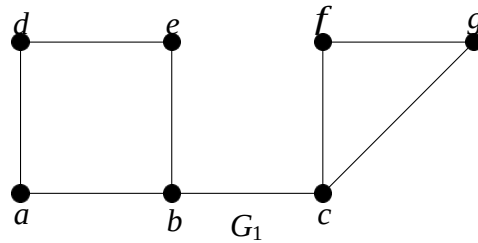
4. Find the complement of the following graph.



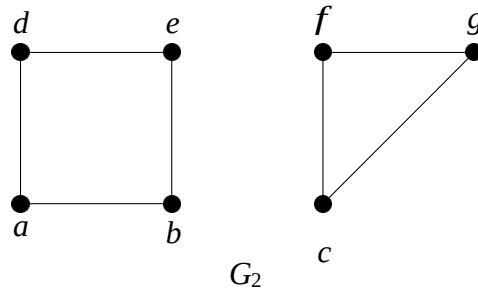
Practical no. 3: Connected Graphs

1. Determine whether the following graphs are connected or not using Maxima software.

1)

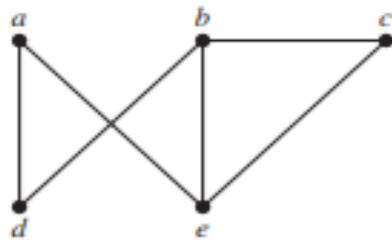


2)

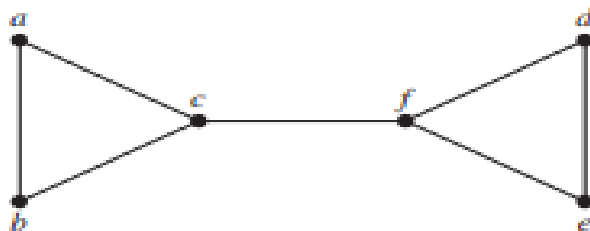


2. Find the edge connectivity and the vertex connectivity of the graphs in following Figure.

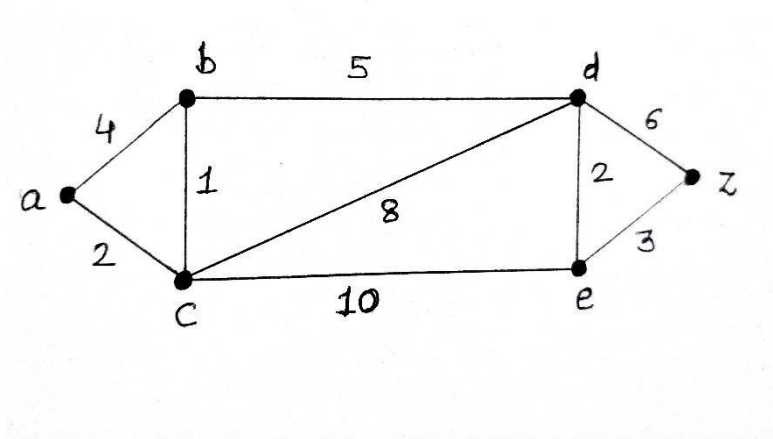
(i)



(ii)



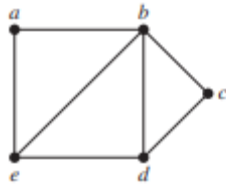
3. Find the shortest path from a to z in the graph below:



Practical no. 4: Trees

1. By using Maxima find the Hamilton path and Hamilton cycle in the following graphs:

(i)



(ii) W_7

2. Draw tree on 7 vertices using Maxima software.

3. Find shortest spanning tree for the following graph using Kruskal's algorithm:

